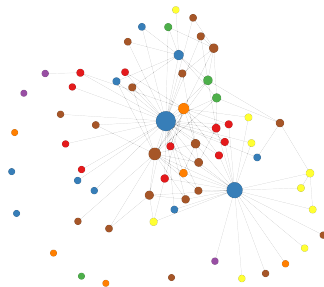


Title of Presentation

Topics



March, 2026

2 Examples

Others

Sequential list display

- Many people use $\text{\LaTeX}$, and many universities have their own Beamer themes

Sequential list display

- Many people use $\text{\LaTeX}$, and many universities have their own Beamer themes
- For Chinese support, use $\text{Xe}\text{\LaTeX}$ as the compiler

List (dynamic) I

- This template is modified from a Tsinghua University project
- Original template: <https://www.overleaf.com/latex/templates/thu-beamer-theme/vwnqzmzndvwyb>
- Minor modifications were made
- Style adjusted to personal preference
- Added small Beamer usage examples
- This is *emphasized text*
- Replaced default fonts with more readable ones

1 Components

Lists

Maths

Tables

Figures

Others

2 Examples

Simple equations

$$V = \frac{4}{3}\pi r^3$$

$$V = \frac{4}{3}\pi r^3$$

Equations

$$E = mc^2$$

$$J(\theta) = \mathbb{E}_{\pi_{\theta}}[G_t] \tag{1}$$

$$Q_{\text{target}} = r + \gamma Q^{\pi}(s', \pi_{\theta}(s') + \epsilon) \tag{2}$$

$$\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma), -c, c) \tag{3}$$

Equation with numbers

$$\begin{aligned}
 A = \lim_{n \rightarrow \infty} \Delta x & \left(a^2 + \left(a^2 + 2a\Delta x + (\Delta x)^2 \right) \right. \\
 & + \left(a^2 + 2 \cdot 2a\Delta x + 2^2 (\Delta x)^2 \right) \\
 & + \left(a^2 + 2 \cdot 3a\Delta x + 3^2 (\Delta x)^2 \right) \\
 & + \dots \\
 & \left. + \left(a^2 + 2 \cdot (n-1)a\Delta x + (n-1)^2 (\Delta x)^2 \right) \right) \\
 & = \frac{1}{3} (b^3 - a^3) \quad (4)
 \end{aligned}$$

Algorithms

Algorithm 1: My algorithm.

Input : G, s

▷ Graph G and a source node s

Output: All nodes reachable from s

```

1   $Q \leftarrow \emptyset$ 
2   $V \leftarrow \emptyset$ 
3  enqueue( $Q, s$ )
4  append( $V, s$ )
5  while  $Q \neq \emptyset$  do
6       $v \leftarrow \textit{dequeue}(Q)$ 
7      print( $v$ )
8      foreach  $u \in N(v)$  do
9          if  $u \notin V$  then
10             append( $V, u$ )
11             enqueue( $Q, u$ )

```

▷ Create an empty queue
▷ Create an empty set of visited

① Components

Lists

Maths

Tables

Figures

Others

② Examples

Table

Table 1: My table.

Representation	Complexity
Adjacency matrix	$O(V ^2)$
Adjacency list	$O(V + E)$
Edge list	$O(E)$
Incidence matrix	$O(V E)$

Table (custom)

Table 2: My table.

Representation	Advantages	Disadvantages
Adjacency matrix	Simple structure, constant-time edge lookup.	High memory usage for large sparse graphs.
Adjacency list	Memory efficient for sparse graphs, easy to iterate over neighbors.	Edge existence check may be slower.
Edge list	Very simple representation, useful for algorithms that process edges sequentially.	Inefficient for neighbor queries.
Incidence matrix	Useful in some theoretical formulations and algebraic graph analysis.	High memory usage and rarely used in practical implementations.

Table (custom)

Table 3: My table.

IA	Tasks
AI learns patterns	AI automates tasks
AI processes data	AI supports decisions
AI powers assistants	AI transforms industries
AI evolves	AI predicts

1 Components

Lists

Maths

Tables

Figures

Others

2 Examples

Figure

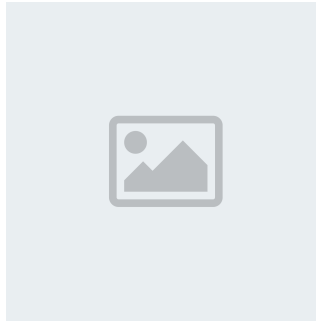
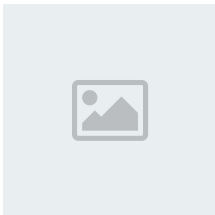


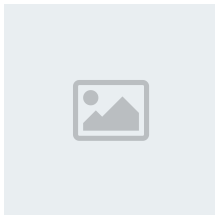
Figure 1: My image.

Source: Author, 2024.

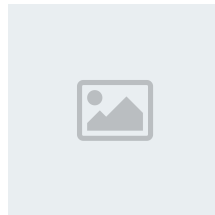
Multiple figures 1



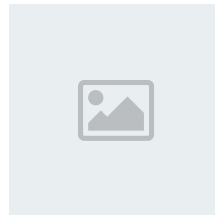
(a) My image.



(b) My image.



(c) My image.



(d) My image.

Figure 2: My image.

Source: Author, 2024.

1 Components

Lists

Maths

Tables

Figures

Others

2 Examples

Blocks 1

Info

General content

Warning

Important message

Example

Example content

Definition

Example definition

Example

Example content

Blocks 2

Theorem (Pythagoras)

$$a^2 + b^2 = c^2$$

Corollary

$$x + y = y + x$$

Proof.

$$\omega + \phi = \epsilon$$



Frame (two columns)

Se enfoca solo en:

- Números (notas)
- Operaciones matemáticas

Se ignora:

- Nombres de cursos
- Detalles del estudiante

Frame (two columns (list of list))

- **Graph traversal**

- Breadth-First Search
- Depth-First Search

- **Shortest path algorithms**

- Dijkstra
- Floyd–Warshall
- Bellman-Ford

Example 6

In this lab, students will apply basic programming concepts to solve problems, develop coding skills, and practice writing and debugging efficient solutions.

- Requirements
 - Python
 - Numpy
 - Pandas
 - NetworkX
 - Matplotlib
- Topics
 - Create graphs
 - Graphs visualization
- Lab
 - [▶ G Open in Google Colab](#)

Example 7

This section introduces key equations to model and analyze the problem.

Main equation

$$J(\theta) = \mathbb{E}_{\pi_\theta}[G_t]$$

Supplementary equations

$$Q_{\text{target}} = r + \gamma Q^\pi(s', \pi_\theta(s')) + \epsilon \quad (8)$$

$$\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma), -c, c) \quad (9)$$

Example 9

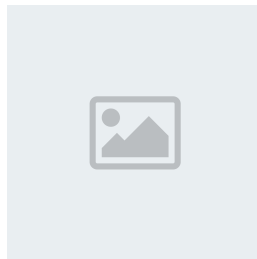
The following table summarizes the key information and results, providing a clear overview of the main variables and their relationships.

Table 4: My table.

Representation	Complexity
Adjacency matrix	$O(V ^2)$
Adjacency list	$O(V + E)$
Edge list	$O(E)$
Incidence matrix	$O(V E)$

The following table summarizes the key information and results, providing a clear overview of the main variables and their relationships.

Representation	Complexity
Adjacency matrix	$O(V ^2)$
Adjacency list	$O(V + E)$
Edge list	$O(E)$
Incidence matrix	$O(V E)$



Source: Author, 2024.

Example 14



Figure 11: My image.

Source: Author, 2024.

Example 15

Descomposición

Consiste en **dividir** el problema en sub problemas o en partes más pequeñas (unidades lógicas de menor complejidad); luego, solucionar cada sub problema independientemente. Finalmente, las soluciones a los subproblemas se integran para resolver el problema original completo.

Ejemplo: Sistema para el cálculo de promedio de notas.

Dividir el problema en:

- 1 Ingresar notas
- 2 Calcular suma
- 3 Calcular promedio
- 4 Mostrar resultado

Example 18

Secuenciales

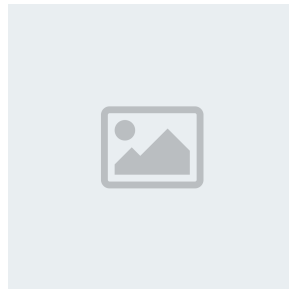
En este tipo de estructura las instrucciones se realizan o se ejecutan una después de la otra.



(a) My image.
Source: Author, 2024.



(b) My image.
Source: Author, 2024.



(c) My image.
Source: Author, 2024.

Figure 12: My image.

Source: Author, 2024.

Example 19

Table 6: Node alignment (*secoSanMartin*).

ID 1	ID 2
13	13
19	252
27	378
31	1148
38	38
43	273
49	72
50	50
51	366
53	241
73	270
...	...
1825	1726

Table 7: Node filtered **133/2072** (*secoSanMartin*).

ID	Rt	Mz	Metabolite name
13	6.10	281.05	Unknown
38	6.16	182.10	Unknown
50	6.20	354.10	Unknown
128	6.47	119.20	Unknown
184	6.76	341.14	Unknown
261	7.23	371.06	Unknown
272	7.29	327.01	Unknown
314	7.61	57.02	DECANE; EI-B; MS
336	7.77	53.96	Unknown
339	7.81	328.01	Unknown
400	8.31	109.08	CITRONELLYL ACETATE...
402	8.31	109.13	BETA-HOMOCYCLOCITRAL...
407	8.37	401.11	Unknown

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